CUSTOMER RETENTION ANALYTICS

Customer ChurnSense

AI System Using Machine Learning

DATA

MODEL

DECISION

Historical customer information as an early churn signal

Gursimran Kaur
Roll No. 02490202024 | BCA 5A
Software Training Report, BCA Vth Semester
Sri Guru Teg Bahadur Institute of Management and Information Technology

Coordinator: Ms. Nehu Gumber
HOD (BCA): Dr. Raj Kumar

Earlier churn signals can create time for customer review

DECISION WINDOW

01 Customer history
Customer profile, relationship, billing and service records

02 Risk prediction
The classifier assigns a likely stay or churn label

03 Customer review
A business team can examine the account context

04 Illustrative action
Engagement, service recovery or a suitable offer

AVAILABLE INFORMATION

PROFILE
Gender and age

RELATIONSHIP
Tenure

BILLING
Monthly and total charges

CONTRACT & SERVICE
Contract type, internet service and related attributes

BINARY TARGET

0 Likely to stay

1 Likely to churn

~1,000
static records described in the synopsis

Boundary
The output supports review. It does not explain why a customer leaves.

Source: Master synopsis, sections A-D, H-I

The documented workflow separates model training from future prediction

TRAINING PATH

1 Inspect	2 Prepare	3 Encode	4 Split	5 Scale	6 Train	7 Evaluate
shape, info, describe	fill missing values	convert categories	80% train, 20% test	fit on training data	Logistic Regression	metrics on test data
data.info()	*fillna()*	*LabelEncoder()*	*train_test_split()*	*StandardScaler()*	*model.fit()*	*classification metrics*

REUSE PATH

Persisted artefacts	New customer record	Transform	Predict
Saved model and scaler, as documented	Same fields, order and category treatment	Reuse the saved preprocessing logic	Return likely stay or churn

CONTROL POINTS

Fit scaling on training data
Reuse transformations for test and new records

Keep test data independent
Prevent evaluation data from influencing learning

Preserve the input schema
Maintain feature order and categorical mappings

Source: Master synopsis, sections D-G and J

Logistic Regression produces the class, while error analysis explains its meaning

MODEL LOGIC

Customer features

Tenure, charges, contract and service attributes

Weighted combination

$p = 1 / (1 + e^{-z})$

WHY THIS BASELINE

It fits a binary classification task and remains comparatively interpretable.

ERROR MATRIX

Actual / predicted	STAY	CHURN
ACTUAL STAY	**Correct stay** No alert needed	**False alert** Unnecessary review
ACTUAL CHURN	**Missed churner** Risk remains unseen	**Correct alert** Review can begin

METRIC INTERPRETATION

Accuracy	Share of all predictions that are correct
Precision	Reliability of churn alerts
Recall	Coverage of actual churners
F1	Balance of precision and recall
Support	Records evaluated in each class

The master synopsis does not report numerical model results.

Classification shows association, not the cause of churn.

Source: Master synopsis, sections E-G; Scikit-learn documentation for metric concepts

A churn prediction becomes useful only when it triggers a disciplined customer review

ILLUSTRATIVE BUSINESS WORKFLOW

1 **Prediction**

Likely stay or churn

2 **Account review**

Check customer context

3 **Response**

Engagement, service recovery or offer

4 **Outcome capture**

Record what happened

DECISION DISCIPLINE

MODEL OUTPUT

A prediction prioritises attention. It does not prescribe an offer.

HUMAN REVIEW

Account context determines whether intervention is appropriate.

ACTION

Engagement and service recovery remain illustrative business responses.

LEARNING LOOP

Recorded outcomes can later support monitoring and retraining.

The current project demonstrates prediction logic. Business impact still requires measured outcomes.

Source: Master synopsis, sections H-J; business workflow shown as an illustrative extension

Production readiness depends on stronger data, broader validation and live integration

CAPABILITY GAP

DIMENSION	DOCUMENTED STATE	WHY IT MATTERS	IMPROVEMENT PATH
Data	Small, static dataset; limited attributes	May not generalise to a different customer base	Refresh data and add complaints or usage patterns
Model	Logistic Regression; Label Encoding	One baseline cannot establish the best approach	Compare other classifiers; use One-Hot Encoding where appropriate
Validation	80:20 hold-out evaluation	One split provides limited evidence of stability	Add cross-validation and hyperparameter tuning
Delivery	Offline data; no live CRM connection	Predictions remain separate from daily operations	Package an application, then integrate refreshed data

DELIVERY SEQUENCE

1 **Strengthen validation** — 2 **Package prediction** — 3 **Integrate refreshed data**

Source: Master synopsis, sections H-J

Code

Paste two or three code screenshots into the editable frames below.

CODE SCREENSHOT 1

Paste image here

CODE SCREENSHOT 2

Paste image here

CODE SCREENSHOT 3

Paste image here

Suggested coverage: preprocessing, model training and prediction output

CERTIFICATE

Certificate

Paste the participant certificate in the frame.

Keep the full certificate visible.
Use Fit rather than Crop.

CERTIFICATE IMAGE

Paste image here

Thank you